Alpine Climate Data Challenge

8 Days to complete & polish everything
22 feb - 3 mar

Need someone good on the :
- modeling of data (predictive models) → use palantir (clean / analyze data) model development & data exploration in SoBig Data
- focused on the front-end deployment (interactive dashboards) → In foundry Visualization & solution - building

Virtual research

SoBig Data → normal environment (load notebook, activate ur models) → upload data to develop the climate model in specific region. (geographical areas / geospatial)
↳ nothing different from Colab (just more powerful)
↳ can use R (python, PySpark, PyTorch, Tensor Flow are pre-installed)

1. Search open dataset
2. clean the data
3. Develop & fine-tune a climate model

- datasets
- raw data
- media files

Analysis of climate and environmental datasets sourced from open sources :
- Copernicus
- Era 5
- APPA Piemonte
- Météo France
- NOAA

Outputs :
- Local climate predictive models based on historical data and IPCC scenarios.
- Interactive dashboards for visualizing climate trends and future simulations
- Monitoring and mitigation solutions supported by AI and data science

Advanced analytical tools for climate data processing :
- Pandas
- Scikit - learn
- Tensor Flow
- Advanced climate models (WRF)

Avoid bias → make accurate predictions
↓
To achieve over 70 % accuracy on test data

Train / validate models

AIP = FOUNDRY + GEN AI

FOCUS ON CLIMATE DATA ANALYSIS IN THE SUSA VALLEY AND MAURIENNE VALLEY, BETWEEN ITALY & FRANCE (WESTERN ALPS)
- DATA ANALYSIS
- PREDICTIVE MODELING
- DEVELOPMENT OF DATA VISUALIZATION SOLUTIONS → PROVIDE REAL ADDED VALUE TO THE AREA → HELP MONITOR

MITIGATE THE CLIMATE CHANGE IMPACT, CONCERNED BY THE NEW LYON - TURIN RAILWAY LANE

1. DATA PROCESSING & MODEL DEVELOPMENT → SoBig Data
2. VISUALIZATION & PROTOTYPE → FOUNDRY

KEY INSIGHTS ON FUTURE CLIMATE IMPACTS

↳ Analyze concrete data to develop climate forecasts and simulations.

1. Rising Temperatures and Heatwaves

Effects : More days above 35°C, increased energy demand, public health risks.

Example : Turin may experience a +3°C temperature rise by 2040, increasing blackout risks due to grid overload.

2. Declining Snow and Ice

Effects : Reduced water resources, impacts on winter tourism, lower hydroelectric capacity.

Example : Chamonix could lose 40 % of its ski slopes by 2040.

3. Increased Floods and Landslides

Effects : Greater hydrogeological risks, damage to infrastructure and transportation.

Example : In 2022, a landslide blocked the Turin - Lyon railway for three weeks.

4. Disappearance of Alpine Species

Effects : Ecosystem disruptions, biodiversity loss.

Example : 50 % of Alpine butterfly species could vanish by 2100.

5. Water Stress and Droughts

Effects : Reduced water reserves for agriculture, human consumption, and hydroelectric power.

Example : The Po River reached its lowest level in 70 years in 2022.

6. Increase in Wildfires

Effects : Forest loss, higher CO_2 emissions.

Example : Wildfires in France and Italy have increased by 400 % compared to the previous decade.

7. Energy Sector Impacts

Effects : Lower hydroelectric output, rising energy costs, summer blackouts.

Example : In 2023, France reduced nuclear power production by 20 % due to low river water levels.

8. Damage to Transport Infrastructure

Effects : Rail and road disruptions, need for resilient infrastructure investments.

Example : Extreme heat deformed rail tracks in Switzerland.

9. Changes in Alpine Tourism

Effects : Decline in winter tourism, need for economic diversification.

Example : Ski resorts below 1,500 m may become unsustainable by 2050.

(10) Changes in Agriculture and Food Production

Effects : Shifts in planting and harvesting cycles, reduced wine quality.

Example : The piedmont grape harvest now occurs 2-3 weeks earlier than 30 years ago.

Insight #1 - Rising Temperatures and Heatwaves

Problem : Hotter summers and their impact on people and the environment.

Temperatures in Alpine regions are rising at nearly twice the global average → Research letters

- More heatwaves : Temperatures exceeding 35°C in historically cool areas.
- Increased water stress : Higher evaporation rates and reduced water resources.
- Impact on biodiversity : Alpine species struggle to adapt and migrate to higher altitudes.

What to do :

- Analyze historical temperature series (NOAA, Copernicus, APPA Piemonte, météo France)
- Develop a temperature prediction model up to 2050 using LSTM or ARIMA.
- Create an interactive map of the most vulnerable areas. (dashboard with Leaflet or mapbox)
- Simulate the impact of heatwaves on economic activities (winter tourism, agriculture)

Case study : The Urban Heat Island Mapping project revealed that Turin will experience a +3°C temperature rise by 2040 in urban areas, increasing the risk of power blackouts due to air conditioning overloads

Insight #2 - Declining Snowfall and Glacier Retreat

Problem : By 2050, snow cover in the Alps could decrease by 30 - 50 % due to global warming.

- Water resources : Less snow = less water available for agriculture and human consumption.
- Winter tourism : Fewer skiable snow days = loss of millions of euros for ski resorts
- Increased hydrogeological risk : Faster glacier melt raises the risk of landslides.

What to do :

- Use copernicus and NOAA data to analyze 30 - year snowfall trends
- Create a predictive model with Random Forest or XGBoost to estimate future snow cover.
- Map water risk areas using geospatial analysis (GIS, GeoPandas)
- Simulate economic scenarios for ski resorts and propose adaptation strategies (artificial snowmaking, summer tourism)

Case study : A météo france study found that chamonix will lose 40 % of its ski slopes by 2040, forcing many resorts to either close or shift to summer tourism

Insight #3 – Increased Extreme Weather Events : Floods and Landslides

Problem : More intense rainfall and hydrogeological risks

The Alps are experiencing more frequent extreme weather events like cloud bursts, landslides, and mudflows. According to the IPCC, extreme precipitation will increase by 15 – 25 % by 2050.

- More floods : Soils unable to absorb water = urban and rural flooding
- More landslides : Permafrost thaw destabilizes mountain slopes, increasing rockfalls.
- Infrastructure damage : railways, roads, and tunnels are more vulnerable to interruptions and costly repairs.

What to do :

- Analyze météo france and ARPA piemonte data to identify periods of extreme rainfall.
- Build a landslide risk map using geological data and GIS models.
- Simulate the economic and social impact of extreme events on key infrastructure (e.g. railway tunnels)
- Propose mitigation solutions : flood retention basins, smart urban drainage systems.

Case study : In 2022, a landslide blocked the Turin – Lyon railway for 3 weeks, causing millions of euros in losses. A machine – learning – based early warning system could help prevent such events.

Insight #4 – Changes in Alpine Biodiversity

Problem : Loss of Alpine species and ecosystem disruption

Climate change is altering Alpine ecosystems, forcing many species to migrate to higher altitudes or face extinction.

- Reduction of Alpine grasslands : Vegetation zones shifting 500+ meters upward.
- Decline of iconic species : Ermine and rock ptarmigan are losing 30 % of their habitats.
- Higher wildfire risk : Hotter temperatures and drier snow increase summer fire risks.

What to do :

- Analyze Copernicus Land Monitoring Service data to track vegetation shifts.
- Develop a predictive model on habitat loss for vulnerable Alpine species.
- Propose conservation strategies, such as natural climate refuges or ecological corridors.

Case study : A university of Innsbruck study predicts that 50 % of Alpine butterfly species could disappear by 2100 without intervention.